# Tryptophan

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## Identity Sheet

| Property | Value |
|---|---|
| **Name** | Tryptophan |
| **Other names** | L-Tryptophan, Trp, W |
| **Chemical formula** | $C_{11}H_{12}N_2O_2$ |
| **Molecular weight** | 204.23 g/mol |
| **CAS number** | 73-22-3 |
| **PubChem CID** | 6305 |
| **Appearance** | White crystalline powder |
| **pKa** | 2.38 (COOH), 9.39 (NH3+) |
| **Solubility** | ~11.4 g/L at 25 C |

**Structural formula:**

```
NH
/ |
H-C=C C-CH2-CH(NH2)-COOH
| | | |
H-C C=C
|
H (indole ring with alanine side chain)
```

Tryptophan is an essential aromatic amino acid and the only amino acid with an **indole ring**, giving it unique fluorescence properties (Ex 280 nm / Em 348 nm). It is a precursor to serotonin, melatonin, niacin (vitamin B3), and kynurenine. Its intrinsic fluorescence makes it the dominant fluorophore in protein UV fluorescence. In urine, tryptophan and its metabolites (indoxyl sulfate, kynurenine, indole-3-acetic acid) serve as markers of protein metabolism, gut microbiome activity, and various diseases.

### Molecules Not to Be Confused With

| Molecule | Formula | MW (g/mol) | Key Difference |
|---|---|---|---|
| **Serotonin (5-HT)** | $C_{10}H_{12}N_2O$ | 176.21 | Tryptophan metabolite; neurotransmitter |
| **Kynurenine** | $C_{10}H_{12}N_2O_3$ | 208.21 | Major tryptophan metabolite via IDO/TDO pathway |
| **Indoxyl sulfate** | $C_8H_6NO_4S^-$ | 212.20 | Gut bacterial metabolite of tryptophan; uraemic toxin |

| Molecule | Formula | MW (g/mol) | Key Difference |
|---|---|---|---|
| **5-HIAA** | $C_{11}H_{11}NO_3$ | 191.18 | Serotonin metabolite; elevated in carcinoid tumours |
| **Tyrosine** | $C_9H_{11}NO_3$ | 181.19 | Another aromatic amino acid; hydroxyphenyl ring; different fluorescence (Ex 275/Em 303 nm) |

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## Medical Information

### Origin

***Endogenous*** Tryptophan is an **essential amino acid** – it must be obtained from diet. It is the least abundant amino acid in the body. It is metabolised via three main pathways: 1. **Kynurenine pathway** (~95% of tryptophan catabolism): Trp -> N-formylkynurenine -> kynurenine -> 3-hydroxykynurenine -> quinolinic acid -> NAD+ (niacin synthesis). Key enzymes: IDO (immune-activated), TDO (liver). 2. **Serotonin pathway** (~1-2%): Trp -> 5-HTP (tryptophan hydroxylase) -> serotonin -> melatonin. 3. **Gut microbial pathway** (~4-6%): Trp -> indole -> indoxyl -> indoxyl sulfate (liver conjugation).

***Exogenous*** Dietary sources: turkey, chicken, milk, cheese, eggs, fish, nuts, seeds, tofu, chocolate. RDA: ~250-425 mg/day (varies by body weight). Typical intake: 200-500 mg/day.

### Primary & Secondary Biological Roles

**Primary role:** - **Protein synthesis:** Essential amino acid for all proteins. - **NAD+ precursor:** Via kynurenine pathway (niacin equivalent).

**Secondary roles:** - **Serotonin/melatonin precursor:** Neurotransmitter and sleep hormone synthesis. - **Immune modulation:** IDO-mediated tryptophan depletion regulates T-cell responses. - **Gut-brain axis:** Microbial tryptophan metabolites (indoles) signal via AhR receptor. - **Autofluorescence marker:** Dominant protein fluorophore at 280/348 nm.

### Catabolism and Elimination Pathway

- ~95% metabolised via the kynurenine pathway (liver and immune cells).
- ~1-2% converted to serotonin.
- ~4-6% metabolised by gut bacteria to indoles.
- Urinary excretion: Trp itself (<50 mg/day) plus metabolites (kynurenine, kynurenic acid, xanthurenic acid, 3-hydroxyanthranilic acid, quinolinic acid, 5-HIAA, indoxyl sulfate).
- Free tryptophan is filtered and ~98% reabsorbed (neutral amino acid transporter).

### Expression in Humans

***Normal Levels***

| Compartment | Reference Range |
|---|---|
| **Plasma tryptophan (total)** | 50-80 umol/L |
| **Plasma tryptophan (free)** | 5-15 umol/L (~10% of total, rest albumin-bound) |
| **Urinary tryptophan** | <50 mg/day (~245 umol/day) |

| Compartment | Reference Range |
|---|---|
| **Urinary 5-HIAA** | 2-8 mg/day (serotonin metabolite) |

***Factors Influencing Levels*** **Increased urinary tryptophan:** - Hartnup disease (defective neutral amino acid transporter; aminoaciduria) - Carcinoid syndrome (5-HIAA elevated, not Trp itself) - Tryptophan supplements - Fanconi syndrome

**Decreased urinary tryptophan:** - Low protein diet - Increased IDO activity (infection, inflammation, cancer) - Vitamin B6 deficiency (impaired kynurenine pathway)

***Associated Pathologies***

| Condition | Tryptophan Pattern | Key Symptoms |
|---|---|---|
| **Hartnup disease** | Massive tryptophanuria + other neutral aminoacids | Pellagra-like rash, cerebellar ataxia; autosomal recessive |
| **Carcinoid syndrome** | Low plasma Trp (diverted to serotonin); high 5-HIAA | Flushing, diarrhoea, wheezing; prevalence ~2/100,000 |
| **CKD (uraemia)** | Elevated indoxyl sulfate (Trp metabolite) | Cardiovascular toxicity, progression of CKD |
| **Depression** | Low plasma free Trp | Mood disorders (serotonin depletion hypothesis) |

### Presence in Urine

**Should it be normally present?** In **small amounts** – free tryptophan is normally present in urine at <50 mg/day. Its metabolites (kynurenine, indoxyl sulfate, 5-HIAA) are also normally excreted.

**Normal urinary levels:** <50 mg/day free Trp; metabolites variable.

**Form in urine:** Free **L-tryptophan** (zwitterionic form at urine pH).

**Solubility:** ~11.4 g/L at 25 C. No crystallisation issues.

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## Detection in Urine

### Available Clinical Assays

1. **HPLC with fluorescence detection:**
   - **Principle:** Reversed-phase C18 HPLC; tryptophan fluorescence detected directly.
   - **Detection: Ex 280 nm / Em 348 nm** (intrinsic indole fluorescence).
   - **LOD:** ~0.05 umol/L (~10 ug/L).
   - **Advantages:** Specific, sensitive, can separate Trp from metabolites.
2. **Amino acid analyser (ion-exchange + ninhydrin):**
   - **Principle:** Ion-exchange chromatography; post-column ninhydrin derivatisation.
   - **Detection:** 570 nm (ninhydrin, primary amines); 440 nm (proline/hydroxyproline).
   - **LOD:** ~1 umol/L.
3. **LC-MS/MS:**
   - **Principle:** RP-HPLC + tandem MS.
   - **Detection:** m/z 205->188 (or 146).

- **LOD:** ~0.01 umol/L.
- **Advantages:** Definitive; multiplex with metabolites.

## Optimal Urine Type for Measurement

- **24-hour urine** for total excretion (aminoaciduria workup).
- Acidify with HCl (stabilises amino acids).
- Freeze at -20 C if delay.

## Actual Gold Standard

**HPLC-fluorescence (Ex 280 / Em 348 nm)** or **LC-MS/MS** for tryptophan and metabolite profiling.

## Interferences in Measurement

| Interference | Effect | Mechanism |
|---|---|---|
| **Protein fluorescence** | Spectral overlap | Tryptophan residues in urinary proteins |
| **Tyrosine** | Partial spectral overlap | Ex 275/Em 303 nm; overlap at excitation |
| **Bacterial metabolism** | Decreased Trp | Bacteria convert Trp to indole |
| **Light** | Photodegradation | Indole ring photosensitive |

## Research Detection Methods

### *Spectroscopy Detection (UV-Vis / NIR)*

- **UV absorption:** Tryptophan absorbs at **280 nm** (epsilon ~5,500 M-1cm-1, indole ring) and **220 nm**. LOD: ~5 umol/L by direct UV (interfered by proteins).
- **NIR:** No useful absorption.

### *Fluorescence Detection*

- **Intrinsic indole fluorescence: Ex 280 nm / Em 348 nm.** This is tryptophan's defining optical property. Quantum yield ~13% (free in water).
  - LOD: ~0.05 umol/L in clean solutions; ~1-5 umol/L in urine (matrix interference).
  - Two-photon excitation: Ex 560 nm.
  - FLIM: Trp lifetime ~3.1 ns (free); variable when protein-bound.
- **EEM (Excitation-Emission Matrix):** Multi-wavelength scanning resolves Trp from tyrosine, NADH, riboflavin.
  - LOD: ~1 umol/L with chemometric deconvolution.
- **Sample prep:** Dilution; filtration for removing proteins.

### *Raman Detection*

| Peak (cm-1) | Assignment |
|---|---|
| **~760** | Indole ring breathing |
| **~880** | Indole N-H deformation |
| **~1010** | Indole C-H in-plane bend |
| **~1340** | Fermi resonance doublet |
| **~1550** | Indole C=C stretch (W3 mode) |

| Peak (cm-1) | Assignment |
|---|---|
| **~1620** | Indole ring |

- **Resonance Raman (Ex 229 nm, deep UV):** Selectively enhances aromatic amino acid modes. LOD: ~10 umol/L.
- SERS: LOD ~0.1-1 umol/L with Ag substrates.

#### *FTIR Detection*

| Band (cm-1) | Assignment | Notes |
|---|---|---|
| **~1660** | C=O stretch (carboxylate) | Overlaps with amide I |
| **~1580** | Indole C=C | – |
| **~740** | Indole ring | More specific |

- Not practical for urinary Trp at typical concentrations.

#### *Voltammetry Detection*

- Tryptophan oxidation: indole ring oxidised at **+0.7-0.9 V vs Ag/AgCl** on carbon electrodes.
- LOD: ~0.1-1 umol/L on modified electrodes (graphene, MIPs, boron-doped diamond).
- Interference: uric acid, ascorbic acid at similar potentials.
- DPV provides best selectivity.

### Other Detection Technologies

1. **Capillary electrophoresis-LIF:** LOD ~0.01 umol/L with UV laser excitation at 266 nm.
2. **Paper-based Ehrlich test:** p-DMAB reacts with indole ring -> violet colour. LOD: ~50 umol/L. Semi-quantitative.
3. **Biosensors (tryptophanase):** Enzyme converts Trp -> indole + pyruvate + NH3. Indole detected optically or electrochemically.

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## Summary Table: Detection Methods Comparison

| Method | LOD | Key Parameter | Sample Prep | Strengths |
|---|---|---|---|---|
| **HPLC-fluorescence** | ~0.05 umol/L | Ex 280/Em 348 nm | None | Gold standard |
| **LC-MS/MS** | ~0.01 umol/L | m/z 205->188 | Extraction | Definitive |
| **Direct fluorescence** | ~1-5 umol/L | Ex 280/Em 348 nm | Dilution | Simple |
| **UV Raman (229 nm)** | ~10 umol/L | 760, 1550 cm-1 | None | Amino acid specific |
| **SERS** | ~0.1 umol/L | 760 cm-1 | Ag NPs | Ultra-sensitive |
| **DPV** | ~0.1 umol/L | +0.8 V | Buffer | Real-time |
| **CE-LIF** | ~0.01 umol/L | Ex 266 nm | None | Separation + detection |

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## References


1. PubChem - L-Tryptophan, CID 6305. https://pubchem.ncbi.nlm.nih.gov/compound/6305
2. StatPearls - Hartnup Disease. https://www.ncbi.nlm.nih.gov/books/NBK482220/
3. StatPearls - Carcinoid Syndrome. https://www.ncbi.nlm.nih.gov/books/NBK448096/
4. PMC - Tryptophan metabolism in health and disease (kynurenine pathway).